CloudFolks HUB
Cloud Computing

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1. Introduction to Cloud Computing

Definition:

Cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing. It enables users to access computing power, storage, and applications without the need for on-site infrastructure.

Components:

- **Server:** The physical or virtual machine that hosts applications and services.
- **Server Rack:** The framework that houses multiple servers in an organized manner.
- **Cooling Systems:** Essential for maintaining optimal operating temperatures for servers.
- **Redundancy:** Ensures reliability and uptime by duplicating critical components and systems.

Key Characteristics:

- **Self-Service Provisioning:** Users can provision resources as needed without human intervention.
- **Elasticity:** Resources can be scaled up or down dynamically to meet changing demands.
- **Pay Per Use:** Users only pay for the resources they consume, offering cost efficiency.
- **Workload Resilience:** Robust infrastructure that ensures high availability and disaster recovery.
- **Migration Flexibility:** Easy movement of applications and data across different cloud environments.

2. Types of Cloud

1. **Private Cloud:**
   - **Description:** Exclusively used by one business or organization, often hosted in on-site data centers.

- **Pros:** Offers more flexibility and improved security.
- **Cons:** Limited scalability compared to public cloud options.

2. **Public Cloud:**
   - **Description:** Resources such as servers and storage are owned and operated by third-party providers like AWS, Azure, and GCP, and delivered over the internet.
   - **Pros:** Lower costs, no maintenance requirements, high scalability, and reliability.
   - **Cons:** Potential concerns about data security and compliance.
3. **Hybrid Cloud:**
   - **Description:** Combines private and public cloud infrastructures, allowing data and applications to be shared between them.
   - **Pros:** Offers control, flexibility, cost-effectiveness, and ease of use.
   - **Cons:** Can be complex to manage and integrate.

3. Cloud Service Models

1. **Infrastructure as a Service (IaaS):**
   - **Components:** Virtualization, servers, storage, networking.
   - **Examples:** AWS EC2, Azure VM.
   - **Advantages:** Highly flexible and dynamic, with pay-as-you-go pricing and automated deployment. Reduces management workload.
   - **Disadvantages:** Data security risks in multitenant environments, potential vendor outages, and requires team training.
   - **Use Cases:** Ideal for startups with limited infrastructure budgets, rapidly growing companies, and large enterprises needing infrastructure control.
2. **Platform as a Service (PaaS):**
   - **Includes:** Operating system, middleware, and runtime.
   - **Examples:** AWS Elastic Beanstalk, AWS RDS, AWS S3.
   - **Advantages:** Scalable, available, and multi-tenant, speeding up development and reducing coding needs. Cost-effective with easy hybrid cloud migration.
   - **Disadvantages:** Security issues, compatibility limitations, and vendor dependency.
   - **Use Cases:** Suitable for collaborative development, use of external vendors, and custom application creation.
3. **Software as a Service (SaaS):**
   - **Description:** Complete products managed by providers, including data and applications.
   - **Examples:** Google Apps, Office 365.
   - **Advantages:** No hardware or setup costs, automated upgrades, and device compatibility. Enables location-independent access and easy customization.
   - **Disadvantages:** Less control over the environment, limited solution range, and dependency on internet connectivity.
   - **Use Cases:** Optimal for quick deployment, short-term collaborations, temporary applications, and web/mobile app needs.

4. **AWS Global Cloud Infrastructure**

5. **AWS Overview:**
   - AWS offers a range of services across IaaS, PaaS, and SaaS models and pioneered the pay-as-you-go model since its inception in 2006.
6. **Region:**
   - A collection of AWS resources available in a specific geographic area. Each region is isolated and independent, consisting of multiple Availability Zones (AZs).
7. **Availability Zones (AZs):**
   - Comprising one or more data centers, AZs offer high bandwidth, low-latency network interconnections and are physically separated by many kilometers, yet remain within 100 km.
8. **Edge Locations:**
   - Act as endpoints for caching content and are more numerous than regions, with over 210+ edge locations globally. These enhance the delivery of content to users worldwide.